

Selectime experiment guideline

1 Question and scope

Test whether validation can select useful Chronos-2 input choices without parameter training. Preserve Adaptime’s 90 dataset/frequency/term tasks and its four dataset exclusions. Compare univariate, multivariate, past-only square-root-magnitude/sign augmentation, top- K for $K = 1, 5, 10, 15, 20$, and both task-level and item/variant-level validation-selected forecasts. The expanded candidates and controls are unevaluated; initial $K \leq 15$ Chronos-2 evidence is summarized separately in the results recap.

2 Protocol

The official TIME test length and horizon H define the unchanged test windows. The default validation length is $H + (\lfloor T_{\text{test}}/H \rfloor - 1)s_{\text{val}}$, with cadence s_{val} inherited from Adaptime and no reserved training interval. Truncate validation at the series beginning if necessary. Model contexts use observed history up to 8192 steps for Chronos-2 or 2048 by default for Bolt and TS-ICL; retrieval uses its configured fixed lookback. Apply the inherited task-specific alignment period and datastore stride, and require complete maximum- K support for covariate forecasts. Direct-horizon transfer requires only one independently searched neighbor. Datastore caps are optional and disabled by default.

Do not send a validation reference to a backbone when its complete available target history contains no finite observation. Such a row contributes no validation loss. If no usable validation dates remain, select univariate input. This rule does not remove any official test cell.

Validation retrieval ends strictly at validation start; test retrieval expands to the pre-test interval after selection. Neighbor futures are finite, wholly before the applicable boundary, and available before the query. Use instance-normalized Euclidean lookback distance and rescale retrieved pasts and futures to query units with past-only statistics. All-series scope is default; exact same-series scope remains an explicit ablation.

3 Selection and randomness

Select all candidate methods by per-date MSSE and the paired bootstrap one-standard-error rule. Use 1000 replications by default. The automatic block length is the larger of rounded $N^{1/3}$ and $\lceil H/s_{\text{val}} \rceil$, capped at $N - 1$. Prefer univariate, multivariate, self-augmentation, then smaller K within the uncertainty threshold. With no usable dates select univariate; with one date use the observed minimum. One run seed (default zero) controls backbone stochasticity and bootstrap resampling. No component seed exists.

4 Controls and backbone grids

Chronos-2 reports eight input candidates, both original selectors, a binary scope selector, and three soft mixtures: scope, top-5 covariate, and nearest neighbor direct horizon (14 methods). TS-ICL excludes multivariate and both scope controls (11 methods). Bolt reports only univariate vanilla and the nearest-horizon mixture (two methods). The intermediate rescaled neighbor horizon is not a reported method or original-selector candidate.

The binary scope selector uses the task-level bootstrap rule over only the two vanilla scopes. Each mixture freezes one scalar task weight from eligible paired validation item/variant/window MSSE wins. Ties contribute one half; alternative fallback rows and non-finite losses are excluded. With w wins in n rows use $p = (1 + w)/(2 + n)$ and output $(1 - p)\hat{y}_{\text{uni}} + p\hat{y}_{\text{alt}}$. No usable validation rows gives $p = 0$. The horizon alternative is the neighbor’s observed historical continuation scaled with lookback-only statistics, without backbone inference. All calibrations precede test and persist their weights/support in artifacts.

5 Execution and artifacts

Hydra configuration is `src/timebench/conf/experiment.yaml`. Dataset settings are `src/timebench/config/datasets.yaml`. The main entry point is `src/scripts/run_selectime.py`. Use `scripts/submit_experiment.sh` for Chronos-2, `scripts/submit_ts_icl.sh` for TS-ICL, and `scripts/submit_chronos_bolt.sh` for Bolt, passing the execution surface. TS-ICL requires `tsicl==0.2.1` in the user-prepared host environment. The standalone selectors are `select_per_task.py` and `select_per_variate.py` in the same directory. They require completed validation forecasts. Scheduler fronts source the common `src/slurm/` workflow, which runs preparation, validation retrieval and forecasting, both selectors, expanded test retrieval and forecasting, selected-prediction assembly, evaluation, and report in order.

Use an already prepared saved-Arrow dataset and a local checkpoint: `chronos2/`, `chronos-bolt-base/`, or `tsicl/tsicl-v1.ckpt` below the configured weights root. The selected Seasonal Naive grid is a prerequisite for test forecasting. All project products live under `outputs/selectime/<backbone>/`; logs under `logs/`. Every stage allocates a schema-1 configuration-addressed manifest before work. Reuse exact completed runs and restart interrupted tasks from their beginning. Reports retain consumed manifests, metric coverage, population dispersion, fallbacks and validation selection counts. Selector and mixture independent inference latency remains unmeasured. Calibration weights and alternatives are included in comparison reports. Forecast stages finalize canonical vanilla after its successful scheduler step before forecasting the other candidates. Report configuration policies are `error/distinct/latest/average`; repeat policies are `selected/latest/distinct/average`. The default uses the current exact configuration and selected repeat. Explicit filters and distinct labels retain nested producer settings. Average repeat task statistics within each exact configuration before averaging configurations; retain every consumed manifest.

6 Verification and evidence

Lightweight scientific regression source exists under `src/tests/`. Synthetic and static regression checks cover chronological datastore boundaries, aligned retrieval, normalization, query-scale covariates, bootstrap selection, the exact 90-task configuration, scheduler fronts and report policies. They do not exercise the foundation backbone or a cluster environment. Obtain a narrow remote

smoke result before the full study, then analyze complete matched artifacts. Initial $K \leq 15$ evidence does not establish performance under the new $K=20$ eligibility or the added controls/backbones. The historical TS-ICL recovery predates the finite-history validation filter; fresh validation predictions and downstream stages are required.